# JEE–MAIN EXAMINATION – JANUARY 2026

**(HELD ON FRIDAY 23rd JANUARY 2026)** **TIME : 9:00 AM TO 12:00 NOON**

## CHEMISTRY

## TEST PAPER WITH ANSWER

### SECTION-A

**51.** Which of the following statements regarding the energy of the stationary state is **true** in the following one-electron system ?

(1) $-1.09 \times 10^{-18}$ J for second orbit of H atom.

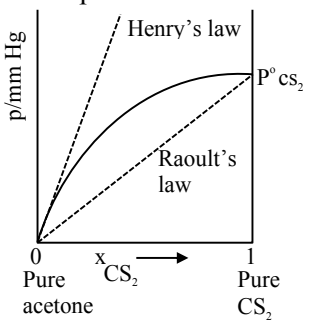
(2) $+2.18 \times 10^{-18}$ J for second orbit of $He^+$ ion

(3) $+8.72 \times 10^{-18}$ J for first orbit of $He^+$ ion

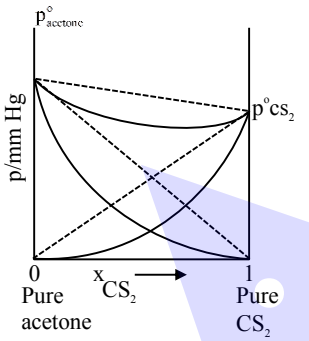
(4) $-2.18 \times 10^{-18}$ J for third orbit of $Li^{2+}$ ion

**52.** Which one of the following graphs accurately represents the plot of partial pressure of $CS_2$ vs its mole fraction in a mixture of acetone and $CS_2$ at constant temperature ?

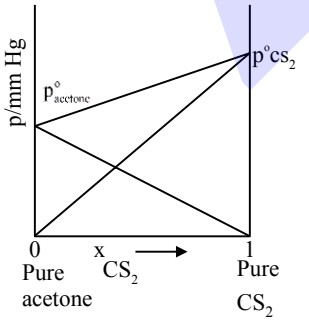
(1)



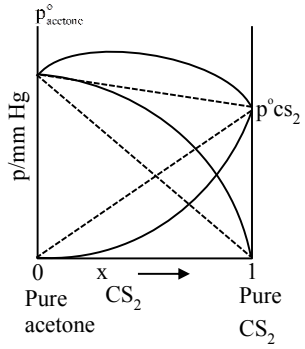

(2)




(3)




(4)

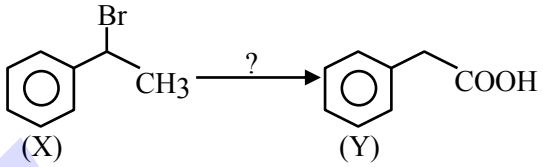



**53.** The correct trend in the first ionization enthalpies of the elements in the 3rd period of periodic table is:

(1) Al < Si < S < P < Cl (2) Al < S < P < Si < Cl

(3) Si < S < Al < P < Cl (4) S < Si < Al < P < Cl

**54.**




The correct sequence of reagents for the above conversion of X to Y is :

(1) (i) NaOH (aq) (ii) Jones reagent (iii) $H_3O^+$

(2) $B_2H_6/H_2O_2$ (ii) NaOEt (iii) Jones reagent

(3) (i) Jones reagent (ii) NaOEt (iii) Hot $KMnO_4/KOH$

(4) (i) NaOEt (ii) $B_2H_6/H_2O_2$ (iii) Jones reagent

**55.** In the given electrochemical cell, Ag(s)|AgCl(s)| $FeCl_2$(aq), $FeCl_3$(aq)|Pt(s) at 298 K, the cell potential ($E_{cell}$) will increase when :

(A) Concentration of $Fe^{2+}$ is increased.

(B) Concentration of $Fe^{3+}$ is decreased

(C) Concentration of $Fe^{2+}$ is decreased

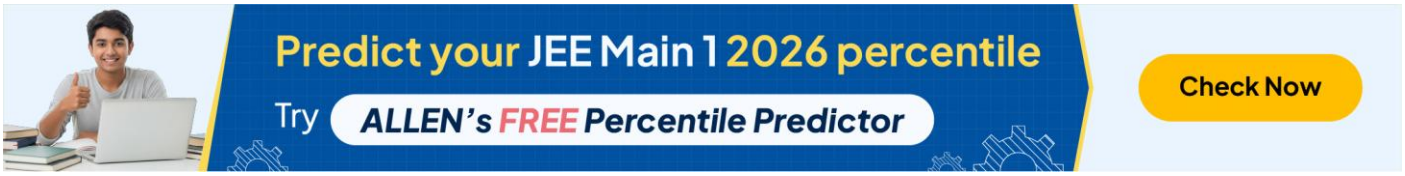
(D) Concentration of $Fe^{3+}$ is increased

(D) Concentration of $Cl^-$ is increased

Choose the **correct** answer from the options given below :

(1) A and B only (2) A and E only

(3) B only (4) C, D and E only

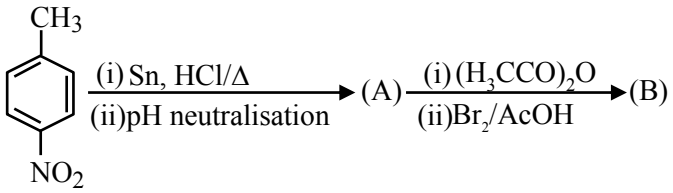


**56.** 'x' is the product which is obtained from propanenitrile and stannous chloride in the presence of hydrochloric acid followed by hydrolysis. 'y' is the product which is obtained from the but-2-ene by the ozonolysis followed by hydrolysis. From the following, which product is not obtained when one mole of 'x' and one mole of 'y' react with each other in the presence of alkali followed by heating ?

(1) 2-Methylbut-2-enal (2) Pent-2-enal

(3) 2-Methylpent-2-enal (4) 3-Methylbut-2-enal

**57.** Consider the following sequence of reactions.




4–Nitrotoluene

Assuming that the reaction proceeds to completion, then 137 mg of 4-nitrotoluene will produce _____mg of B.

(Given molar mass in g $mol^{-1}$ H : 1, C : 12, N :14, O :16, Br : 80)

(1) 301 (2) 146

(3) 228 (4) 208

**58.** A cup of water at 5°C (system) is placed in a microwave oven and the oven is turned on for one minute during which, the water begins to boil. Which of the following option is **true** ?

(1) q = +ve, w = 0, $\Delta U$ = –ve

(2) q = +ve, w = –ve, $\Delta U$ = +ve

(3) q = –ve, w = –ve, $\Delta U$ = –ve

(4) q = +ve, w = –ve, $\Delta U$ = –ve

**59.** Given below are two statements :

**Statement I :** $[CoBr_4]^{2-}$ ion will absorb light of lower energy than $[CoCl_4]^{2-}$ ion.

**Statement II :** In $[CoI_4]^{2-}$ ion, the energy separation between the two set of d-orbitals is more than $[CoCl_4]^{2-}$ ion.

In the light of the above statements, choose the correct answer from the options given below :

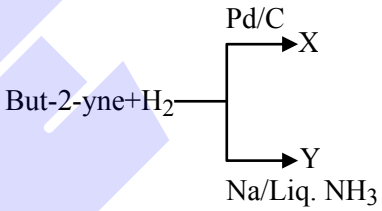
(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is true but **Statement II** is false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are true

**60.** But-2-yne and hydrogen (one mole each) are separately treated with (i) Pd/C and (ii) Na/liq. $NH_3$ to give the products X and Y respectively.




Identify the **incorrect** statements.

A. X and Y are stereoisomers.

B. Dipole moment of X is zero

C. Boiling point of X is higher than Y.

D. X and Y react with $O_3$/Zn + $H_2O$ to give different products.

Choose the **correct** answer from the options given below :

(1) B and C only (2) B and D only

(3) A and B only (4) A and C only

**61.** Given,

(A) n = 5, $m_l = -1$

(B) n = 3, l = 2, $m_l = -1$, $m_s = +\frac{1}{2}$

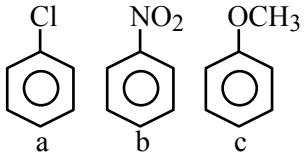
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

(1) 26 and 1 (2) 4 and 1

(3) 2 and 4 (4) 8 and 1



**62.** Consider the following compounds




Arrange these compounds in the increasing order of reactivity with nitrating mixture.

(1) c < a < b (2) b < c < a

(3) c < b < a (4) b < a < c

**63.** The statements that are **incorrect** about the nickel (II) complex of dimethylglyoxime are :

A. It is red in colour

B. It has a high solubility in water at pH = 9

C. The Ni ion has two unpaired d-electrons

D. The N – Ni – N bond angle is almost close to 90°

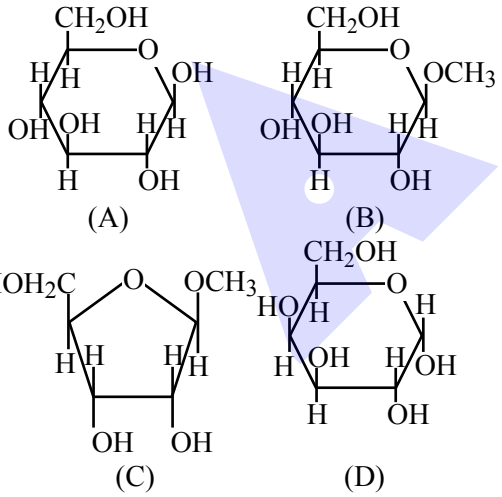
E. The complex contains four five-membered metallacycles (metal containing rings)

Choose the **correct** answer from the options given below :

(1) C and E only (2) A, D and B only

(3) B, C and E only (4) C and D only

**64.** From the given following (A to D) cyclic structures, those which will not react with Tollen's reagent are :




(1) B and D (2) A and D

(3) A and B (4) B and C

**65.** Identify the molecule(X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among $HNO_3$, $H_2SO_4$, $NF_3$ and $O_3$. Choose the **correct** bond angle made by the central atom of the molecule (X).

(1) 120° (2) 107°

(3) 102° (4) 116°

**66.** Match List-I with List-II.

| | **List-I Functional group (detection)** | | **List-II Change observed during detection** |
|---|---|---|---|
| (A) | Unsaturation (Baeyer's test) | (I) | Red colour Appears |
| (B) | Alcoholic group (Ceric ammonium nitrate test) | (II) | Silver mirror appears |
| (C) | Aldehyde group (Tollen's reagent) | (III) | Violet colour appears |
| (D) | Phenolic group ($FeCl_3$ test) | (IV) | Discharge of pink colour |

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-I, C-II, D-III

(4) A-IV, B-III, C-II, D-I



**67.** Given below are two statements :

**Statement-I :** Sublimation is used for the separation and purification of compounds with low melting point.

**Statement-II :** The boiling point of a liquid increases as the external pressure is reduced.

In the light of the above statements, choose the **correct** answer from the options given below :

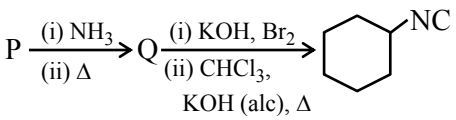
(1) Statement-I is false but Statement-II is true.

(2) Statement-I is true but Statement-II is false.

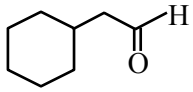
(3) Both Statement-I and Statement-II are true.

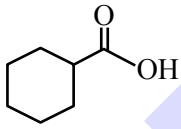
(4) Both Statement-I and Statement-II are false.

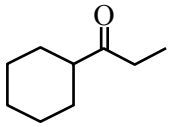
**68.** Compound 'P' undergoes the following sequence of reactions :

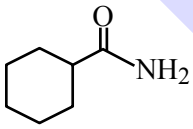



'P' is :

(1) 


(2) 


(3) 


(4) 


**69.** The correct statements from the following are :

(A) Ionic radii of trivalent cations of group 13 elements decreases down the group.

(B) Electronegativity of group 13 elements decreases down the group.

(C) Among the group 13 elements, Boron has highest first ionisation enthalpy.

(D) The trichloride and triiodide of group 13 elements are covalent in nature.

Choose the **correct** answer from the otpions given below :

(1) A and C only (2) A and D only

(3) C and D only (4) B and D only

**70.** Consider the general reaction given below at 400 K

$xA(g) \rightleftharpoons yB(g)$

The values of $K_p$ and $K_c$ are studied under the same condition of temperature but variation in x and y.

(i) $K_p = 85.87$ and $K_c = 2.586$ appropriate units

(ii) $K_p = 0.862$ and $K_c = 28.62$ appropriate units

The value of x and y in (i) and (ii) respectively are:

(1) (i) 3, 1 ; (ii) 3, 1

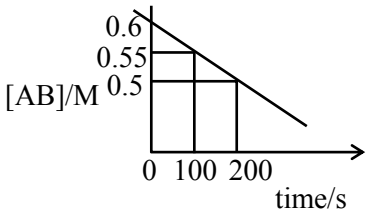
(2) (i) 4, 1 ; (ii) 4, 1

(3) (i) 1, 3 ; (ii) 2, 1

(4) (i) 1, 2 ; (ii) 2, 1

## SECTION-B

**71.** For the thermal decomposition of reaction AB(g), the following is constructed.




The half life of the reaction is 'x' min.

x = _____ min. (Nearest integer)

**72.** For the following gas phase equilibrium reaction at constant temperature,

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

If the total pressure is $\sqrt{3}$ atm and the pressure equilibrium constant ($K_p$) is 9 atm, then the degree of dissociation is given as $(x \times 10^{-2})^{-1/2}$.

The value of x is _____ (Nearest integer)



**73.** x mg of pure HCl was used to make an aqueous solution. 25.0 mL of 0.1M $Ba(OH)_2$ solution is used when the HCl solution was titrated against it. The numerical value of x is _____ $\times 10^{-1}$. (Nearest integer)

Given : Molar mass of HCl and $Ba(OH)_2$ are 36.5 and 171.0 g $mol^{-1}$ respectively.

**74.** Consider all the structural isomers with molecular formula $C_5H_{11}Br$ are separately treated with KOH (aq) to give respective substitution products, without any rearrangement. The number of products which can exhibit optical isomerism from these is _____.

**NTA**

**75.** The crystal field splitting energy of $[Co(oxalate)_3]^{3-}$ complex is 'n' times that of the $[Cr(oxalate)_3]^{3-}$ complex. Here 'n' is _____. [Assume $\Delta_0 >> P$]



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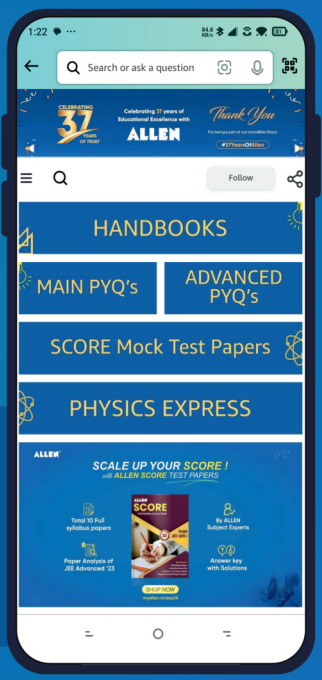
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